

AURORA Conceptual Governance Loop

Observation

Edge telemetry

$O_t = \langle \text{delay, cpu, memory, throughput, fps} \rangle$

Parallel micro-agents

single-responsibility
anomaly checks

Anomaly vector A_t

SLO predicates fired

Diagnosis

Phase 2 — Root-cause analysis

Markov-blanket-bounded BN posterior
 $P(C | O_t)$

Composite ranking $\rho(C)$

impact $\Delta(C)$ · evidence ·
path factor · structural prior

Dual-gate

Gate 1 — Posterior certainty

$P_{\max} \geq \tau_{\text{eff}}$

Gate 2 — VFE bound

$\mathcal{F}_{\text{total}} < \mathcal{F}_{\text{th}}$

Action

Local mitigation

dispatch $\text{do}(a)$ on the edge node

Abstain → fog tier

serialised diagnostic payload

Feedback
fog re-inference
BN re-fit

Edge node → Continuum Bridge → fog tier